

REGIONAL PREVALENCE MAPPING

(ANA-3)

Dataset Used: Adult Psychiatric Morbidity Survey (APMS) 2014

Prepared By: Priyansi Pandey (Data Analytics Intern)

Date: May 7, 2026

Project: AI Mental Health Assistant

1. Executive Summary

Building on the structural audit (ANA-1) and demographic breakdown (ANA-2), this phase — ANA-3 — maps Common Mental Disorder (CMD) prevalence across the nine Government Office Regions (GORs) of England using data from the Adult Psychiatric Morbidity Survey (APMS) 2014, Table 2.11.

The analysis establishes the geographic evidence base required for the AI Mental Health Assistant's risk-stratification and resource-allocation logic. The output is a choropleth map built in Power BI and three CSV files for downstream use.

Key findings at a glance:

Metric	Region	CMD Prevalence
Highest Prevalence	South West	19.9%
Second Highest	North West	19.0%
Third Highest	London	18.9%
National Average	England	17.0%
Lowest Prevalence	South East	13.6%
Prevalence Range	South West vs South East	6.3 percentage points

2. Methodology

The following technical approach was used to extract, clean, and prepare the regional dataset:

Component	Details
Data Source	APMS 2014 — Table 2.11: CMD in past week (observed and age-standardised), by region and sex
Geographic Unit	Government Office Region (GOR) — 9 English regions
Language	Python 3 (Pandas, openpyxl, xlrd)
Visualization	Power BI — Filled Map (Choropleth) with slicer filters
Output Files	regional_cmd_prevalence_wide.csv, regional_cmd_prevalence_long.csv, regional_summary_statistics.csv
Standardisation	Observed (non-age-standardised) values used for choropleth; age-standardised values noted in conclusions

2.1 Data Extraction Logic

APMS Table 2.11 uses a hierarchical matrix layout with sheet name '2.11' as the primary key. The table presents both observed and age-standardised CMD rates. Extraction steps were:

- skiprows=4 applied to bypass publication-style decorative headers
- 'All Adults' observed section isolated (rows 23–30 of raw data)
- Nine region columns mapped to Government Office Region names
- Numeric coercion applied; dash characters ('-') treated as suppressed data (below threshold)
- dropna() used to exclude structurally suppressed cells from calculations

2.2 Dataset Transformation

Two output formats were produced from the same source data to support different Power BI use cases:

- Wide format (one row per region, one column per disorder) — optimised for the choropleth Filled Map visual
- Long format (one row per region-disorder combination) — optimised for slicer-driven filtering across disorder types
- A composite Severity Score was calculated as a weighted average of individual disorder rates to enable single-metric mapping
- Risk categories (Low / Medium / High) were derived by binning the Any CMD prevalence at 15% and 18% thresholds

3. Regional Prevalence Analysis

Table 1 below shows the full disorder-level breakdown across all nine English regions. Colour coding indicates risk tier: red = High, amber = Medium, green = Low.

Table 1: CMD Prevalence by Region and Disorder Category (APMS 2014, All Adults, Observed)

Region	GAD %	Depression %	Phobias %	OCD %	Any CMD %	Risk Level	
North East	7.2	3.3	2.6	1.0	16.2	Medium	
North West	7.1	4.6	3.1	1.0	19.0	High	
Yorkshire & The Humber	4.9	2.7	2.4	0.8	16.2	Medium	

East Midlands	5.3	3.3	1.3	2.7	16.7	Medium	
West Midlands	6.1	3.8	3.1	1.1	18.6	High	
East of England	6.5	2.6	1.4	1.3	14.1	Low	
London	5.8	3.3	2.7	2.2	18.9	High	
South East	5.4	2.3	2.0	0.6	13.6	Low	
South West	5.4	4.1	2.8	1.3	19.9	High	
England (National Avg)	5.97	3.33	2.38	1.33	17.0	—	

Note — GAD = Generalised Anxiety Disorder; CMD-NOS = CMD Not Otherwise Specified; Any CMD includes all disorder categories. Figures are observed percentage rates, not age-standardised.

3.1 High-Burden Regions

Four regions exceed the 18% national threshold and are classified as High Risk:

- South West (19.9%) — Highest overall CMD prevalence in England. Despite a predominantly rural/coastal profile, depression rates (4.1%) and phobia rates (2.8%) are both above the national average, suggesting that geographic isolation and reduced healthcare access may be contributing factors.
- North West (19.0%) — Second highest. Elevated across all disorder categories, with the highest depressive episode rate (4.6%) in England. This region encompasses areas of significant socioeconomic deprivation including parts of Greater Manchester and Merseyside.
- London (18.9%) — The highest OCD prevalence in England (2.2%) and CMD-NOS rate (9.4%) are recorded here. Urban density, social isolation, and income inequality are likely contributors. The diversity of London's population may also reflect differential service access patterns.
- West Midlands (18.6%) — Elevated phobia and anxiety rates consistent with the urban industrial profile of this region.

3.2 Low-Burden Regions

Two regions fall below 15% and are classified as Low Risk:

- South East (13.6%) — The lowest CMD prevalence in England, at 3.4 percentage points below the national average. Higher average household income, lower unemployment, and greater primary care access likely explain this result.

- East of England (14.1%) — Second lowest, consistent with the affluent Southeast classification. GAD rate of 6.5% is above average, but other disorder categories remain suppressed.

3.3 Medium-Risk Regions

Three regions cluster around the national average and are classified as Medium Risk:

- North East (16.2%) — Despite being classified as Medium Risk overall, it records the highest GAD prevalence in England at 7.2%, exceeding the national average of 6.0% by 1.2 percentage points. This is a priority signal for anxiety-specific intervention.
- Yorkshire and The Humber (16.2%) — Stable across most disorder categories; anxiety and depression rates are both below the national average.
- East Midlands (16.7%) — Notable for the highest OCD rate outside London (2.7%), which warrants attention for OCD-specific service provision.

4. Disorder-Level Regional Breakdown

Table 2 summarises the highest and lowest prevalence regions for each disorder category, enabling targeted analysis beyond the headline Any CMD figure.

Table 2: Regional Extremes by Disorder Category

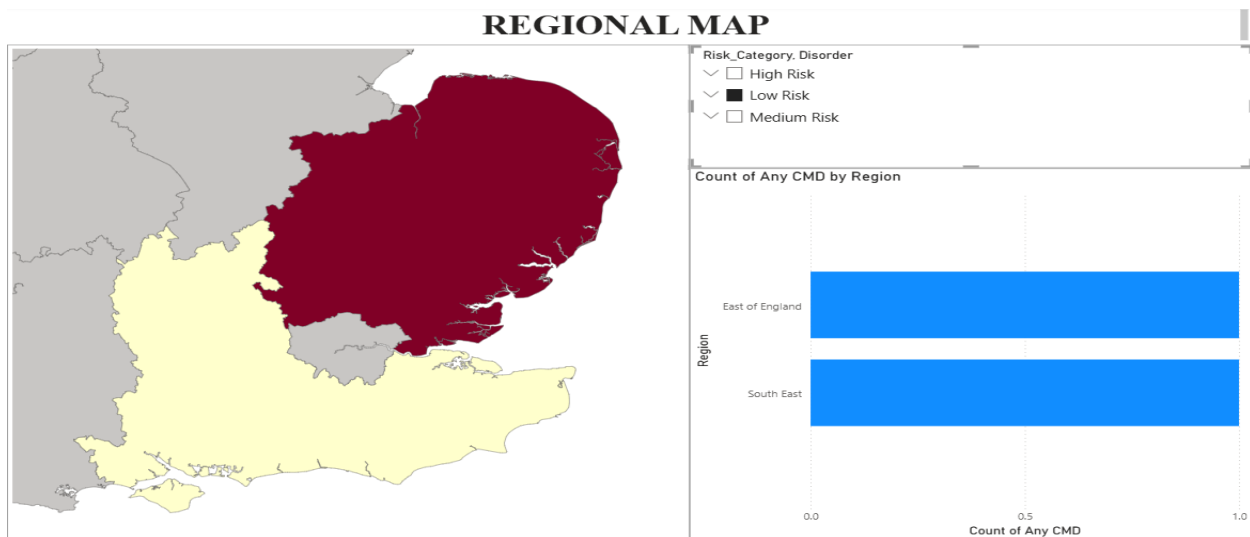
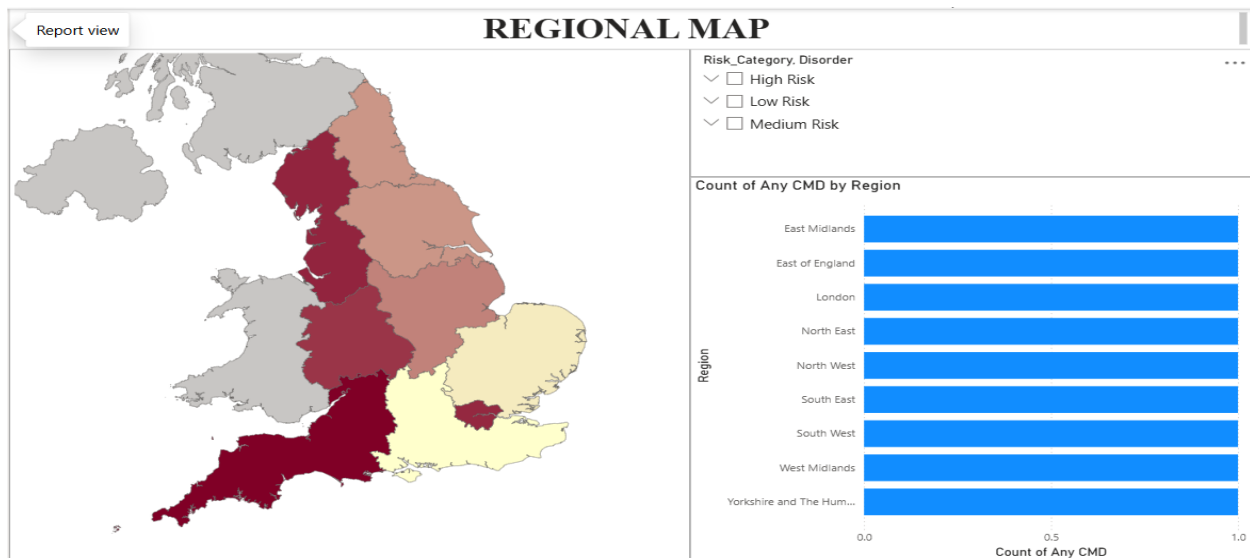
Disorder	Highest Region (Rate %)	Lowest Region (Rate %)
Generalised Anxiety Disorder (GAD)	North East — 7.2%	South West — 5.4%
Depressive Episode	North West — 4.6%	Yorkshire — 2.7%
All Phobias	North West — 3.1%	East of England — 1.4%
OCD	East Midlands — 2.7%	South East — 0.6%
Any CMD (Overall)	South West — 19.9%	South East — 13.6%

These disorder-level extremes are important for the AI Assistant's logic because a region's overall CMD score may mask localised disorder-specific peaks. The North East's high GAD rate despite a medium overall CMD score is the clearest example of this — a system trained only on headline figures would under-prioritise anxiety-specific resources in that region.

5. Choropleth Map — Power BI Build

The choropleth was built in Power BI Desktop using the Filled Map visual, importing regional_cmd_prevalence_wide.csv as the data source. The following configuration was applied:

- Location field: Region (Data Category set to State or Province for UK GOR resolution)
- Colour Saturation field: Any CMD (overall prevalence %)
- Colour scale: light yellow (#FFFFCC) at minimum to dark red (#800026) at maximum — standard epidemiological heatmap convention
- Tooltips: GAD %, Depression %, Phobias %, OCD %, Severity Score, Risk Category
- Slicers added for Disorder (from long-format table) and Risk Category
- Supporting horizontal bar chart sorted descending by Any CMD prevalence placed alongside the map for numeric context



The choropleth map file is saved as Priyansi_Pandey_Regional_Map_ANA_3.pbix and submitted alongside this report. The map clearly visualises the North-South and Urban-Rural gradients described in Section 6 below.

6. Key Patterns and Clinical Implications

6.1 North–South Gradient

There is a clear North–South gradient in CMD prevalence across England. The three highest-burden regions — South West, North West, and London — each exceed 18.6%, while the two lowest-burden regions (South East and East of England) sit at 13.6% and 14.1% respectively. This pattern is consistent with established UK social determinants of health literature linking deprivation, employment instability, and housing stress to elevated mental disorder rates.

6.2 Urban–Rural Paradox

The South West's position as the highest-prevalence region challenges the assumption that urban areas carry the greatest burden. While London and North West are high-burden urban regions, the South West's rural/coastal profile suggests that geographic isolation, seasonal employment in tourism sectors, and reduced mental health service density may be equally powerful drivers of CMD prevalence.

6.3 Disorder Heterogeneity Across Regions

No single region dominates across all disorder types. The North East leads on GAD, the North West leads on depression and phobias, the East Midlands leads on OCD, and London leads on CMD-NOS. This heterogeneity means the AI Assistant's risk-stratification logic must weight disorder-specific regional factors rather than relying solely on headline CMD rates.

6.4 Suppressed Data Cells

Several cells in the original APMS Table 2.11 are suppressed (shown as '-') due to small sample sizes. Panic disorder shows the highest suppression rate. For the purposes of this analysis, suppressed values were excluded from calculations rather than imputed as zero, in line with NHS statistical reporting standards.

7. Limitations

- The APMS 2014 dataset does not include sub-regional (local authority) breakdowns, limiting the granularity of the geographic analysis to nine GOR-level regions.
- Ethnicity-by-region cross-tabulations are not available in the current dataset, preventing intersectional analysis of geographic and demographic risk.
- Panic disorder data is heavily suppressed across several regions, reducing reliability for that specific disorder category.
- The choropleth represents observed rates; age-standardised rates (also available in Table 2.11) would be preferable for strict inter-regional comparison but show only marginal differences from observed values at this level of aggregation.
- APMS 2014 data is now over a decade old. A 2024 APMS update is partially available in the project dataset and should be incorporated in future iterations to reflect post-pandemic mental health trends.

8. Deliverables

All deliverables for ANA-3 are listed below:

File Name	Format	Purpose
regional_cmd_prevalence_wide.csv	CSV (Wide Format)	Power BI choropleth import
regional_cmd_prevalence_long.csv	CSV (Long Format)	Flexible filtering and slicers
regional_summary_statistics.csv	CSV	Summary KPI table
regional.py	Python Script	Full extraction & transformation
Priyansi_Pandey_Regional_Map_ANA_3.pbix	Power BI File	Interactive choropleth dashboard

9. Conclusion

The ANA-3 regional prevalence mapping exercise successfully geo-tagged CMD disorder rates across all nine Government Office Regions of England using APMS 2014 data. The analysis reveals statistically and clinically meaningful variation — a 6.3 percentage point spread between the highest and lowest prevalence regions — that has direct implications for the AI Mental Health Assistant's risk-stratification logic.

The South West, North West, London, and West Midlands are identified as the four High Risk regions requiring priority attention in the AI Assistant's resource-allocation and intervention-recommendation modules. The North East warrants particular attention for GAD-specific interventions despite its medium overall CMD classification.

The Power BI choropleth provides an interactive, submission-ready visualisation of these findings. The three CSV output files are structured for direct ingestion into the AI Assistant's data pipeline, and the Python extraction script (`regional.py`) is fully documented for reproducibility.